# Solar Dynamic Characterization of Fine Temporal Structures in Fixed-Frequency Radio Emissions

*Mayte Alvarez Trimino and Adolfo L. Mendez, Ph.D.*
Miami Dade College, Padron Campus, Miami, FL

## Introduction

- On October 28, 2003, the Sun emitted an intense X17.2 flare, followed by an X10. These were originated from NOAA Active Region 0486, the most significant of solar cycle 23 (Aurass et al., 2006; Cliver et al., 2022; Pick et al., 2005).
- The analyzed radio bursts, part of the 2003 Halloween Solar Storms, reveal energetic flare and CME processes that severely impacted satellites, power grids, and communications on Earth.
- Our analysis identifies spike structures as highly deterministic and chaotic, offering insights into solar radio dynamics and their diagnostic value in solar physics.

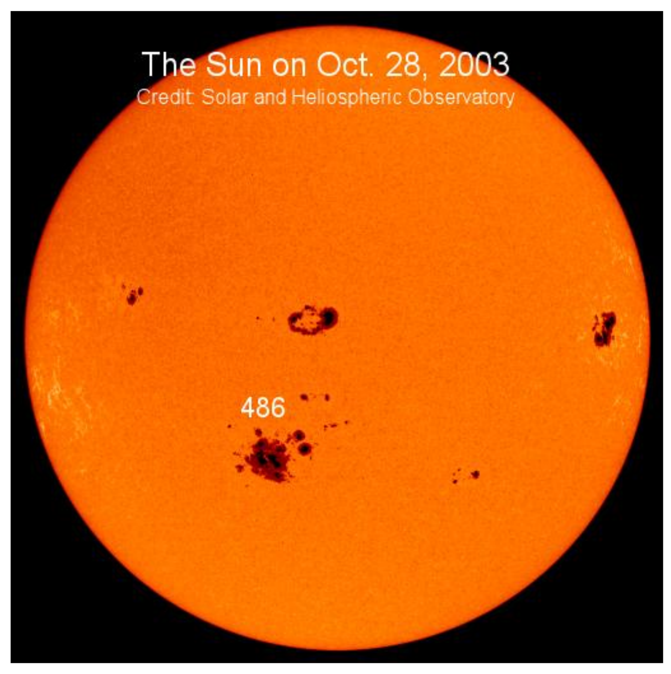


**Fig. 1** Solar image of NOAA Active Region 0486 on October 28, 2003, credit SOHO

## Methodology

- We analyzed high-resolution time series data (1,000 data points per second) recorded by the Trieste Astronomical Observatory at 237 MHz, 327 MHz, and 408 MHz.
- Two optimal 4,000-point intervals (10:00:10-10:00:14 UT, 10:00:14-10:00:18 UT) were selected.

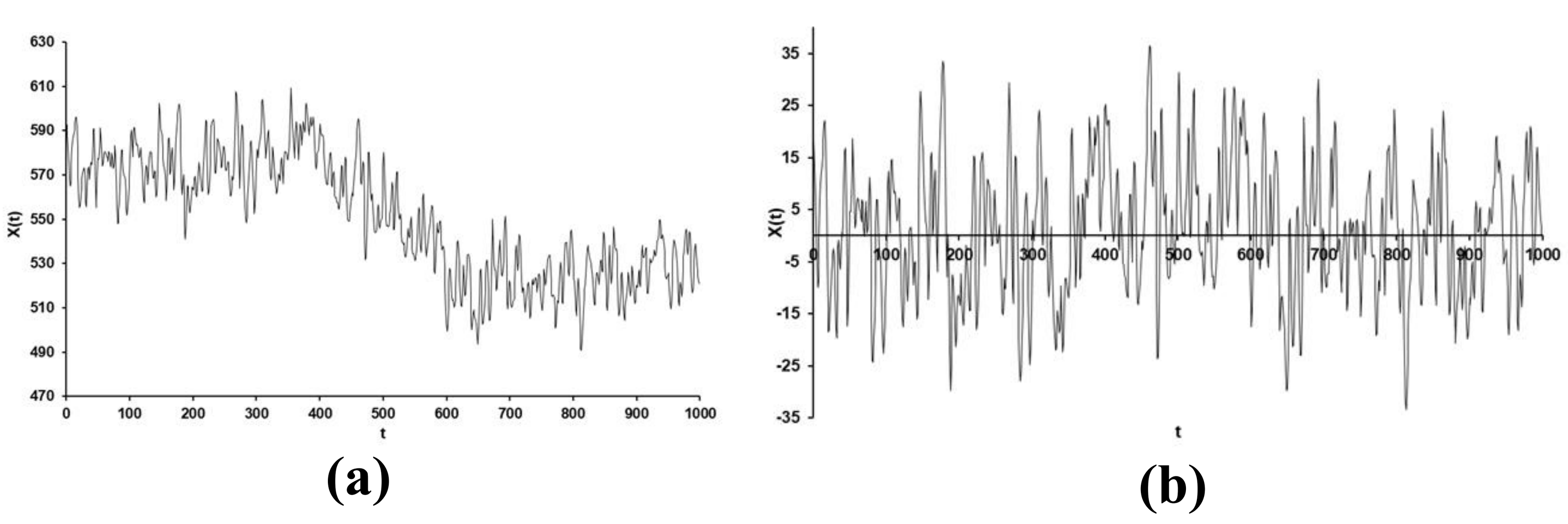


**Fig. 2** Time profile of the fine structure at 237 MHz with a time resolution 1000 measurements per second **(a)** original signal, and **(b)** background-removed signal

- Stationarity was assessed using power spectral density (Welch, 1967) and bandpower stability across windows, following standard recommendations for nonlinear time-series analysis (Kantz & Schreiber, 2004).
- We applied time-delay embedding (Takens, 1981) to reconstruct the phase space using:

$$\vec{X}(t) = [x(t), x(t+\tau), \ldots, x(t+(m-1)\tau)]$$

- Nonlinear Analysis included:
  - Fast Fourier Transfrom (FFT)
  - Lyapunov Exponent (**λ**)
  - Hurst Exponent (**H**)
  - Deterministic Factor (**Δ**)
  - Correlation Dimension (**DC**)
  - Hausdorff Dimension (**DH**)
- All computations were carried out using Excel, Dataplore (TISEAN package), Matjaž Perc's CHAOS software, and MATLAB for nonlinear dynamic measures.

## Results

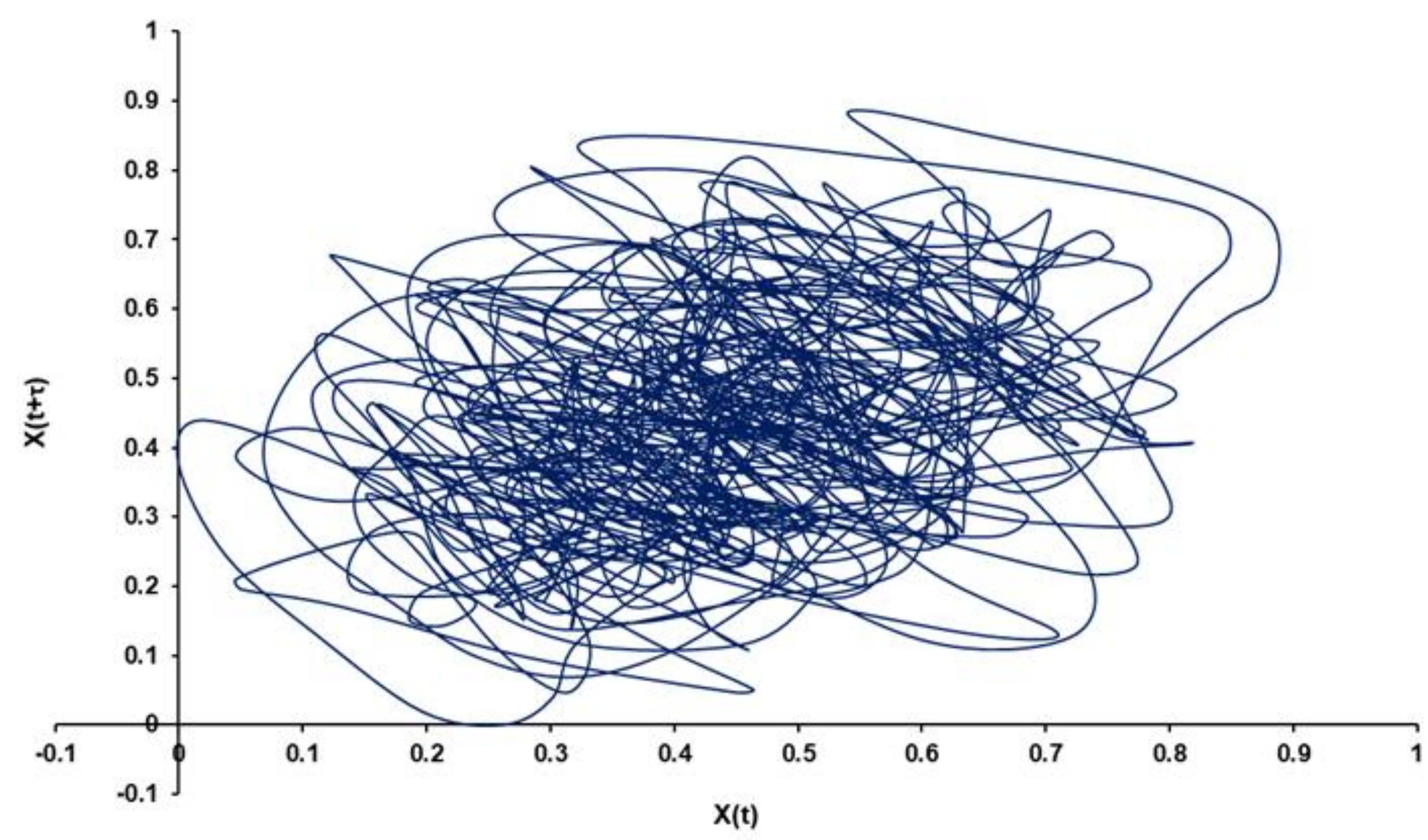


**Fig. 3** 2D state-space was reconstructed at 237 MHz using a time delay τ = 4, determined from the first minimum of average mutual information.

| Frequency | λ | Δ | H | $D_C$ | $D_P$ |
|---|---|---|---|---|---|
| 237 MHz | 0.449 | 0.708 | 0.267 | 6.019±0.114 | 6.793±0.524 |
| 327 MHz | 0.279 | 0.678 | 0.257 | 6.097±0.129 | 6.964±0.558 |
| 408 MHz | 0.560 | 0.623 | 0.263 | 6.055±0.100 | 6.733±0.477 |

**Table 1**. Lyapunov exponent **λ** (Wolf et al., 1985), Deterministic factor **Δ,** Hurst exponent **H**, Correlation dimension $\mathbf{D_C}$, and Hausdorff dimension $\mathbf{D_H}$

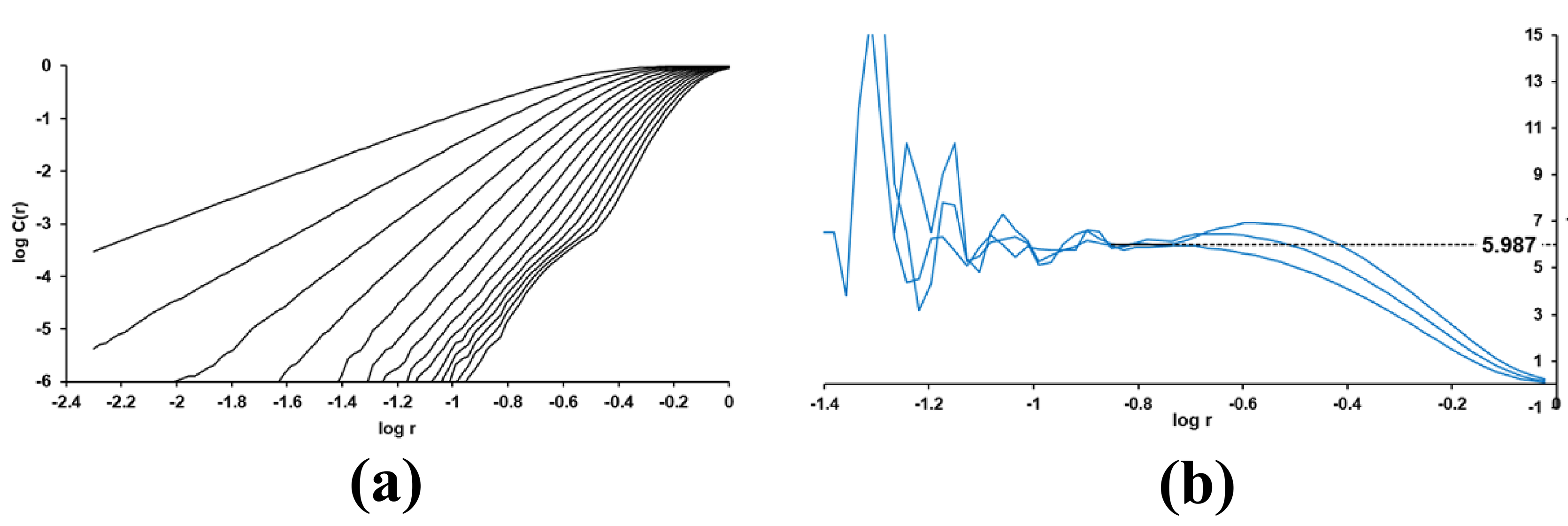


**Figure 4**. **(a)** Correlation integral log [$C_d^{(2)}$] vs. log [r] for the time series at 237 MHz **(b)** Slopes of the linear scaling regions for 237 MHz (Grassberger, 1987).

## Conclusions

- The results suggest that spiky activity represents a **high-dimensional chaotic system**, differing from previously studied structures like fibers and pulsations, likely due to distinct generation mechanisms (Mendez et al., 2013, 2015)
- Future work could explore broader frequency ranges and time intervals, apply advanced methods like multifractal analysis or machine learning, and include data from multiple solar cycles to uncover long-term patterns in radio burst dynamics.

## Additional Info

Scan QR Code
to access References
and Acknowledgments